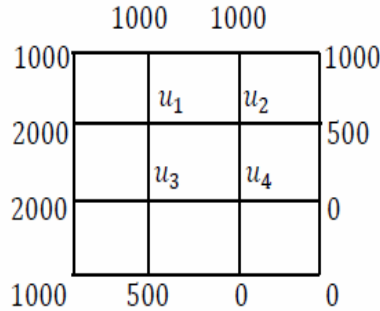


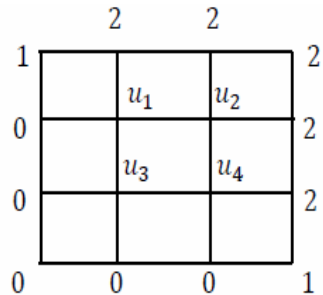
Numerical methods for PDE

Laplace equation

1. Determine the value of $u(x, y)$ satisfying the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ at the pivotal points of a square region with boundary values as shown in the figure. Carry out 3 iterations.



2. Solve $u_{xx} + u_{yy} = 0$ in the following square region with the boundary conditions indicated as shown in the figure. Perform 3 iterations.



3. Solve $u_{xx} + u_{yy} = 0$ in $0 \leq x \leq 3, 0 \leq y \leq 3$ given that $u(x, 0) = 0, u(0, y) = 0, u(x, 3) = x^2$ and $u(3, y) = 3y$. Take $h = k = 1$ and employ Gauss Seidel method to solve this problem and carry out three iterations by considering the value up to 4 decimals.
4. Solve $u_{xx} + u_{yy} = 0$ in $0 \leq x \leq 1, 0 \leq y \leq 1$ given that $u(x, 0) = 0, u(0, y) = 0, u(x, 1) = 6x$ and $u(1, y) = 6y$. Take $h = k = \frac{1}{3}$ and employ Gauss Seidel method to solve this problem. Carry out three iterations by considering the value up to 4 decimals.

Heat Equation

1. Solve the heat equation $\frac{\partial u}{\partial t} = 4 \frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(0, t) = 0 = u(8, t)$ and $u(x, 0) = 4x - \frac{x^2}{2}$ at the points $x = i: i = 0, 1, 2, \dots, 8$ and $t = \frac{1}{8}j: j = 0, 1, 2, \dots, 5$.

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2. Solve the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(x, 0) = \sin \pi x, 0 \leq x \leq 1$ $u(0, t) = u(1, t) = 0$ using Schmidt method. Carry out computations for two levels, taking $h = \frac{1}{3}$ and $k = \frac{1}{36}$.
3. Solve the heat equation $\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(0, t) = 0 = u(4, t)$ and $u(x, 0) = x(4 - x)$. Employ Bendre-Schmidt method with $h = 1$ to find the solution of the equation for $0 \leq t \leq 1$.
4. Solve the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(0, t) = 0 = u(5, t)$ and $u(x, 0) = x^2(25 - x^2)$ by taking $h = 1$ for $0 \leq t \leq 2$ using Bendre-Schmidt method.

Wave equation

1. By using explicit 3 level formula, solve the wave equation $u_{tt} = 4u_{xx}$ subject to the boundary conditions $u(0, t) = u(4, t) = 0, t \geq 0$ and the initial conditions $u(x, 0) = x(4 - x), \frac{\partial u}{\partial t}(x, 0) = 0, 0 \leq x \leq 4$. Take $h = 1$ and $k = \frac{1}{2}$ and compute $u(x, t)$ for $0 \leq t \leq 2$.
2. Solve $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ with the boundary conditions $u(0, t) = u(1, t) = 0, t \geq 0$ and the initial conditions $u(x, 0) = \sin(\pi x), \frac{\partial u}{\partial t}(x, 0) = 0, 0 \leq x \leq 1$ taking $h = k = 0.2$ at $t = 1$.
3. Evaluate the pivotal values of the equation $u_{tt} = 16u_{xx}$ taking $h = 1, k = \frac{1}{4}$ up to $t = 1$. The boundary conditions are $u(0, t) = u(5, t) = 0$ and the initial conditions $u(x, 0) = x^2(5 - x)$ and $\frac{\partial u}{\partial t}(x, 0) = 0$.
4. By using explicit three-level formula, solve the wave equation $4 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$ subject to the conditions $u(0, t) = u(1, t) = 0, t \geq 0, u(x, 0) = \begin{cases} \frac{5x}{3}, & 0 < x \leq \frac{3}{5} \\ \frac{5(1-x)}{2}, & \frac{3}{5} < x < 1 \end{cases}$ and $\frac{\partial u}{\partial t}(x, 0) = 0$. Taking $h = \frac{1}{5}, k = \frac{1}{10}$ carry out computations up to $t = \frac{2}{5}$.

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COMPLEX ANALYSIS

1. Show that the following functions are analytic

(a) $F(z) = x + \sin x \cosh y + i(y + \cos x \sinh y)$

(b) $F(z) = \frac{x-1}{(x-1)^2 + y^2} - i \frac{y}{(x-1)^2 + y^2}$

2. Prove that analytic function with constant modulus is constant.

3. If $f(z)$ is analytic function, show that

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right] |f(z)|^2 = 4|f'(z)|^2$$

4. An electrostatic field in the xy – plane is given by the potential function $\phi = x^2 - y^2$, find stream function.

5. Prove that $u = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic. Find a function v such that $f(z) = u + iv$ is analytic.

6. Find the orthogonal trajectories of the family of curves $r^2 \cos 2\theta - r \cos \theta + 2 = c$

7. In a two dimensional fluid flow, the complex potential is $w(z) = u + iv$ show that velocity potential $u = e^x(x \cos y - y \sin y)$ is harmonic and find its harmonic conjugate.

8. Find the analytic function $f(z)$ whose imaginary part is $e^x(x \sin y + y \cos y)$

9. Obtain the power series expansion of the function $f(z) = \frac{z^2 - 1}{z^2 + 5z + 6}$ in the following regions.

(i) $|z| < 2$

(ii) $2 < |z| < 3$

(iii) $|z| > 3$

10. Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in Laurent's series valid for

(i) $0 < |z + 1| < 2$, (ii) $|z + 1| > 2$

11. Evaluate the following integrals using Cauchy's residue theorem.

(i) $\int_C \frac{1-2z}{z(z-1)(z-2)} dz$, $C: |z| = 1.5$ (ii) $\int_C \frac{z+4}{z^2+2z+5} dz$, $C: |z + 1 - i| = 2$

12. Evaluate the integral $\int_C \frac{1}{z^2(z^2+4)} dz$ where C is the circle $|z - 2i| = 3$ using Cauchy's residue theorem.